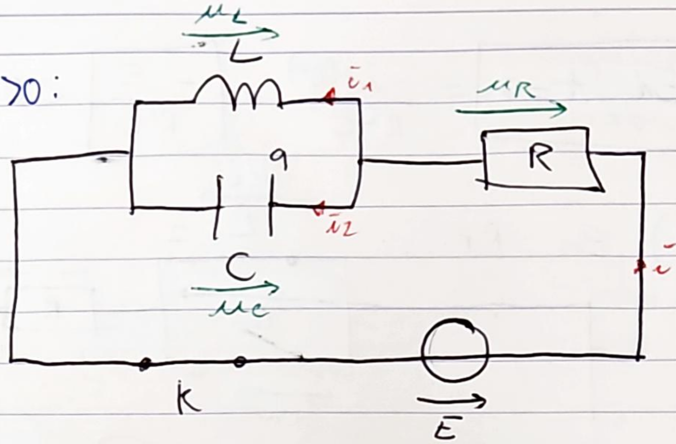
Exercice 6 : Pour $t > 0$:



1) À $t=0^-$: C déchargé : $u_C(0^-)=0 \Leftrightarrow C \equiv$ —

donc L court-circuitée : $i_1(0^-)=0$

À $t=0^+$: les discontinuités | tension aux bornes C : $u_C(0^+)=0$

| courant ~~aux bornes~~ dans L : $\boxed{i_1(0^+)=0}$

LDN : $i = i_1 + i_2$

donc $i(0^+) = i_1(0^+) + i_2(0^+)$

$\Leftrightarrow i(0^+) = i_2(0^+)$

$q(0^+) = u_C(0^+) \times C$

$\Leftrightarrow \boxed{q(0^+) = 0}$

LDM : $E = u_R + u_C$

donc $E = u_R(0^+) + u_C(0^+)$

$\Leftrightarrow E = u_R(0^+)$

Loi d'Ohm (conv récept) : $u_R(0^+) = R\,i(0^+)$

$$\Leftrightarrow \boxed{i(0^+) = \frac{E}{R}}$$

$$\Leftrightarrow \boxed{i_2(0^+) = \frac{E}{R}}$$

Loi d'Ohm (conv récept) : $u_R = Ri$

LDM : $E - u_C = Ri$

$$\Leftrightarrow \frac{1}{R}(E-u_C) = i$$

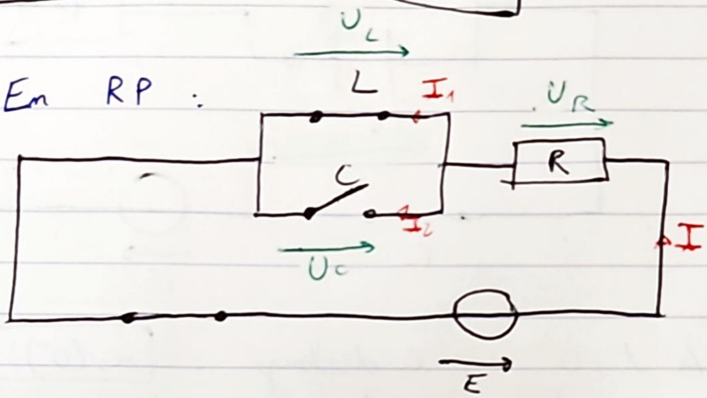
$$\Leftrightarrow \frac{1}{R}\frac{d(E-u_C)}{dt} = \frac{di}{dt}$$

$$\Leftrightarrow -\frac{1}{R}\frac{du_C}{dt} = \frac{di}{dt}$$

$$\Leftrightarrow -\frac{1}{R}\frac{i_2}{C} = \frac{di}{dt}$$

$$\boxed{\left(\frac{di(t)}{dt}\right)_{t=0^+}} = \frac{-i_2(0^+)}{R\cdot C} = \boxed{\frac{-E}{R^2C}} < 0$$

2) En RP :



$C \equiv$ —/— donc $\boxed{I_2 = 0}$

$L \equiv$ —•—• donc $U_L = 0$

LDN : $I = I_1 + I_2 \Leftrightarrow I = I_1$

LDM : $E = U_R + U_L$

$= U_R$

Loi d'Ohm (conv recept) : $U_R = RI$

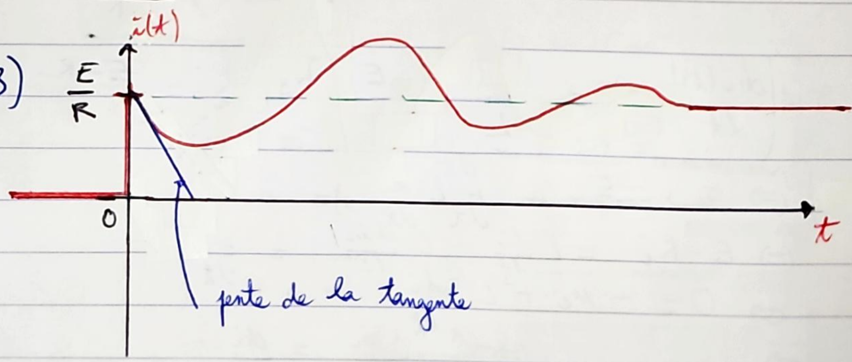
$\Leftrightarrow \boxed{I = \frac{E}{R} = I_1}$

$q_P = U_C \times C$ or $U_L = U_C = 0$

$\Leftrightarrow \boxed{q_P = 0}$

$$\boxed{\left(\frac{di(t)}{dt}\right)_{t=\infty}} = \frac{-I_2}{RC} = \boxed{0}$$

3)



4) LDM : $E = u_R + u_C$

$E = u_R + u_L$

LDN : $i = i_1 + i_2$

Loi d'Ohm (conv récept.) : $u_R = Ri$

Lois courant - tension (conv récept) : $i_2 = C \frac{du_C}{dt}$

$u_L = L \frac{di_1}{dt}$

$E = u_R + u_L$

$\Leftrightarrow E = Ri + L \frac{di_1}{dt}$

$\Leftrightarrow E = Ri + L \frac{di}{dt} - L \frac{di_2}{dt}$

$\Leftrightarrow E = Ri + L \frac{di}{dt} - LC \frac{d^2 u_C}{dt^2}$

$\Leftrightarrow E = Ri + L \frac{di}{dt} - LC \frac{d^2 E - u_R}{dt^2}$

$\Leftrightarrow E = Ri + L \frac{di}{dt} + LC \frac{d^2 u_R}{dt^2}$

$\Leftrightarrow E = Ri + L \frac{di}{dt} + LCR \frac{d^2 i}{dt^2}$

$\Leftrightarrow \frac{E}{LCR} = \frac{d^2 i}{dt^2} + \frac{1}{RC} \frac{di}{dt} + \frac{1}{LC} i$

$\Leftrightarrow \boxed{\omega_0^2 \frac{E}{R} = \frac{d^2 i}{dt^2} + \frac{\omega_0}{Q} \frac{di}{dt} + \omega_0^2 i \quad \text{avec} \quad \omega_0 = \frac{1}{\sqrt{LC}}}$

$$E = u_R + u_L$$

$$\Leftrightarrow E = Ri + L\frac{di_1}{dt}$$

$$\Leftrightarrow E = R(i_1+i_2) + L\frac{di_1}{dt}$$

$$\Leftrightarrow E = Ri_1 + Ri_2 + L\frac{di_1}{dt}$$

$$\Leftrightarrow E = Ri_1 + RC\frac{du_C}{dt} + L\frac{di_1}{dt}$$

$$\Leftrightarrow E = Ri_1 + RC\frac{d(E-u_R)}{dt} + L\frac{di_1}{dt}$$

$$\Leftrightarrow E = Ri_1 + RC\frac{d(E-E+u_L)}{dt} + L\frac{di_1}{dt}$$

$$\Leftrightarrow E = Ri_1 + RCL\frac{d^2 i_1}{dt^2} + L\frac{di_1}{dt}$$

$$\Leftrightarrow \frac{E}{RLC} = \frac{d^2 i_1}{dt^2} + \frac{1}{RC}\frac{di_1}{dt} + \frac{1}{LC} i_1$$

$$\Leftrightarrow \boxed{\omega_0^2 \frac{E}{R} = \frac{d^2 i_1}{dt^2} + \frac{\omega_0}{Q}\frac{di_1}{dt} + \omega_0^2 i_1 \quad \text{avec } \omega_0 = \frac{1}{\sqrt{LC}}}$$